Computer Science & IT



Data Structure & Programming

Stack

Lecture No. 02



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Recap of Previous Lecture



- Topic — stack implementation
- Topic — init , isEmpty
- Topic — push pop
- Topic — LIFO
- Topic

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Stack application Expression

Stack as permutation generator

Slide



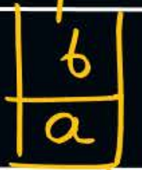
Topic : Stack as Permutation Generator

* we can take character in order, a, b, c
* we are given a stack
* we will push element in given order
* pop & print can happen.

Slide



Topic : Stack as Permutation Generator

a b c	push(a) pop(a) push(b) pop(b) push(c) pop(c)
a c b	push(a) pop push(b) push(c)  pop pop
b a c	push(a) push(b)  pop pop push(c) pop
b c a	push(a) push(b)  pop push(c)  pop, pop
c a b	push(a) push(b) push(c)  pop pop(b)
c b a	push(a) push(b) push(c) pop pop pop

3 distinct alphabet

5 Sequence possible

✓ ✓ ✓ X ✓

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Topic : Stack as Permutation Generator

a, b, c d

push(a)
pop

a b c d ✓
a b d c ✓
a c b d ✓
a c d b ✓
a d b c X
a d c b ✓

b a c d
b a d c
b c a d
b c d a
b d a c
b d c a

c a b d
c a d b
c b d a
c b a d
c d a b
c d b a

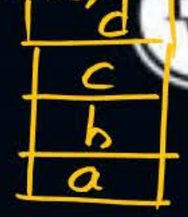
d a b c
d a c b
d b a c
d b c a
d c b a
d c a b

Slide



Topic : Stack as Permutation Generator

push(a) push(b) push(c) push(d) pop



a, b, c d

5	5	3	1
a b c d ✓ (push(a) pop)	b a c d	c a b d	d a b c X
a b d c ✓	b a d c	c a d b	d a c b X
a c b d ✓	b c a d	c b d a	d b a c X
a c d b ✓	b c d a	c b a d	d b c a X
a d b c X	b d a c	c d a b	d c b a ✓
a d c b ✓	b d c a	c d b a	d c a b X

Slide



Topic : Stack as Permutation Generator

5

push(a) push(b) pop

b



pop push(c) push(d)

pop pop



push(a) push(b) pop



push(c) push(d)

b a c d		c a b d
b a d c		c a d b
b c a d		c b d a
b c d a		c b a d
b d a c	✗	c d a b
b d c a	✓	c d b a

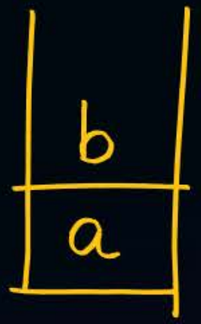
Slide



Topic : Stack as Permutation Generator

push (a) push (b) push (c) pop

c



pop order fixed to b a

c a b d	X
c a d b	X
c b d a	✓
c b a d	✓
c d a b	X
c d b a	✓

3 permutation possible

Slide



Topic : Stack as Permutation Generator

a, b, c, d, with 4 distinct alphabet 14 permutation are possible

with n distinct alphabet No. of permutation possible

represented by catalan Number.

$$C_n = \frac{1}{n+1} {}^{2n}C_n$$

$$C_3 = \frac{1}{4} {}^{6}C_3 \leftarrow \frac{\lfloor 6}{\lfloor 3 \lfloor 3}$$

$$= \frac{1}{4} \frac{\lfloor 6}{\lfloor 3 \lfloor 3}$$

$$= \frac{1}{4} \frac{6 \times 5 \times 4 \lfloor 3}{3 \times 2 \times 1 \times \lfloor 3} = 5$$

Slide



Topic : Stack as Permutation Generator

$C_1 = 1$
$C_2 = 2$
$C_3 = 5$
$C_4 = 14$
$C_5 = 42$
$C_6 = 132$

a, b, c d

push(a)

Slide



Topic : Stack as Permutation Generator

#Q The following sequence of operations is performed on a stack:

PUSH (10), PUSH (20), POP , PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP

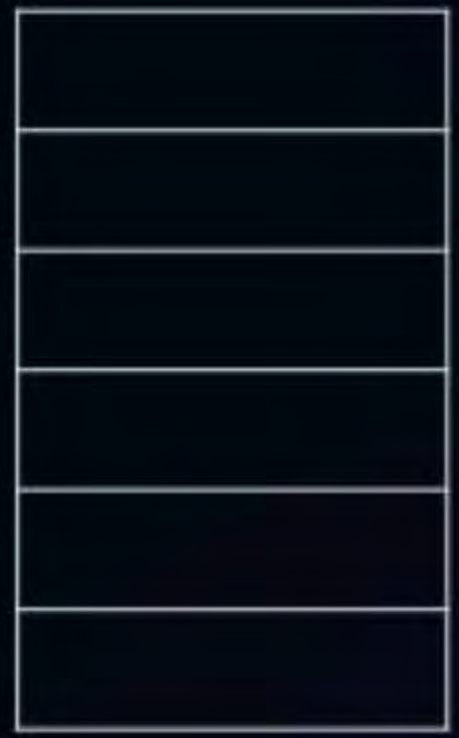
The sequence of values popped out is:

(A) 20, 10, 20, 10, 20

(B) 20, 20, 10, 10, 20

(C) 10, 20, 20, 10, 20

(D) 20, 20, 10, 20, 10





Topic : GATE 2004

A program attempts to generate as many permutations as possible of the string 'abcd' by pushing the characters a, b, c, d in the same order onto a stack, but it may pop off the top character at any time. Which one of the following strings CANNOT be generated using this program?

(A) abcd

(b) dcba

(c) cbad

(d) cabd

push(a) push(b) push(c) push(d)
pop pop pop pop

push(a) push(b) push(c)
pop



Slide

Topic : GATE 1994

Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence $1, 2, 3, 4, 5$ in that order?

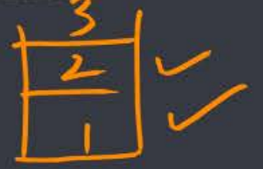
A. $3, 4, 5, 1, 2$ ✗

B. $3, 4, 5, 2, 1$ ✓

C. $1, 5, 2, 3, 4$ ✗

D. $5, 4, 3, 1, 2$ ✗

push(1) push(2) push(3)



push(1) pop push(2) push(3) push(4) push(5)





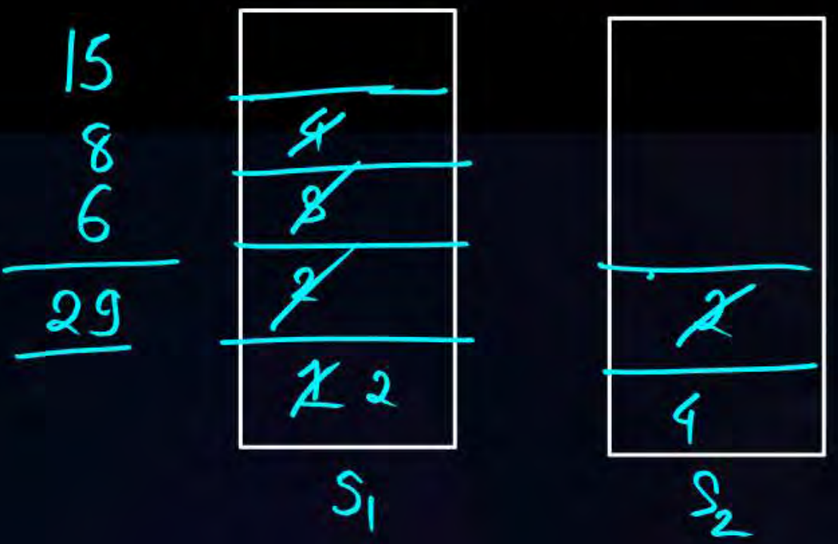
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Question

#Q. Push operation takes 2 units of times, pop takes 1 unit of time, printing an element takes 3 units of time. Suppose S_1, S_2 are 2 stacks where S_1 has elements {1, 2, 3, 4, 5}, 5 is at top and S_2 is empty. The min unit of time required to print the sequence 53142 is_____.

(Note: only push, pop print can be used.)



15
8
6
29

5
pop, print, $X = pop(S_1)$, $push(S_2, X)$

3
pop(S_1), print, $X = pop$ $push(S_2, X)$

2
pop print $X = pop(S_2)$ $push(S_1)$

pop, pop(S) print pop(S_1) print

```
#include <stdio.h>
int *A, stkTop;
int stkFunc (int opcode, int val) {
   static int size=0, stkTop=0;
   switch (opcode) {
      case -1: size = val; break;
      case 0: if (stkTop < size ) A[stkTop++]=val; break;
      default: if (stkTop) return A[--stkTop];
   }
   return -1;
}
int main() {
   int B[20]; A=B; stkTop = -1;
   stkFunc (-1, 10);
   stkFunc (0, 5);
   stkFunc (0, 10);
   printf ("%d\n", stkFunc(1, 0)+ stkFunc(1, 0));
}
```

The output of the program is _____



stkTop = 0 1 2

A = 100 , top = -1

Size = 10

0 < 10

1 < 10

10 + 5

15

Slide

```
#include <stdio.h>
int *A, stkTop;
int stkFunc (int opcode, int val) {
   static int size=0, stkTop=0;
   switch (opcode) {
      case -1: size = val; break;
      case 0: if (stkTop < size ) A[stkTop++]=val; break;
      default: if (stkTop) return A[--stkTop];
   }
   return -1;
}
int main() {
   int B[20]; A=B; stkTop = -1;
   stkFunc (-1, 10);
   stkFunc (0, 5);
   stkFunc (0, 10);
   printf ("%d\n", stkFunc(1, 0)+ stkFunc(1, 0));
}
```

The output of the program is _____



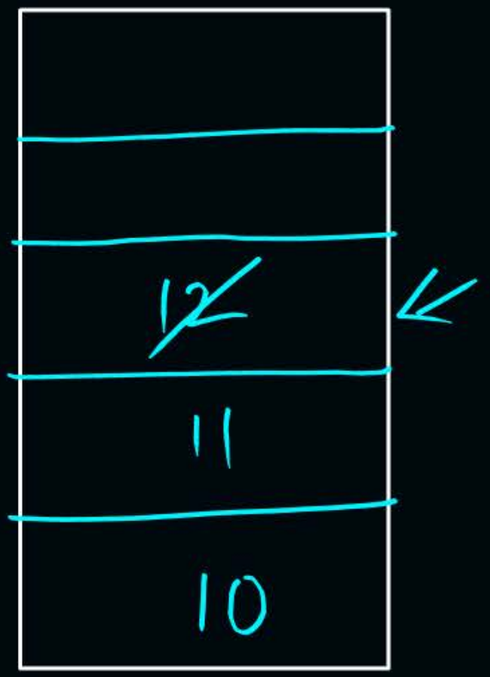
stkTop 0 1 2 1

A = 100

Size = 10 0 < 10

10 + 5 = 15

Slide



pop : Empty underflow

push(10), push(11), push(12)

X = pop()



2 mins Summary

Topic — Stack permutation

Topic — Catalan No

Topic — problems

Topic

Topic

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THANK - YOU

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